

MARCH 18 – 22, 2026

BALTIMORE, MD

#NSBE2026

Advancing S.T.E.M.



2026 ANNUAL CONVENTION

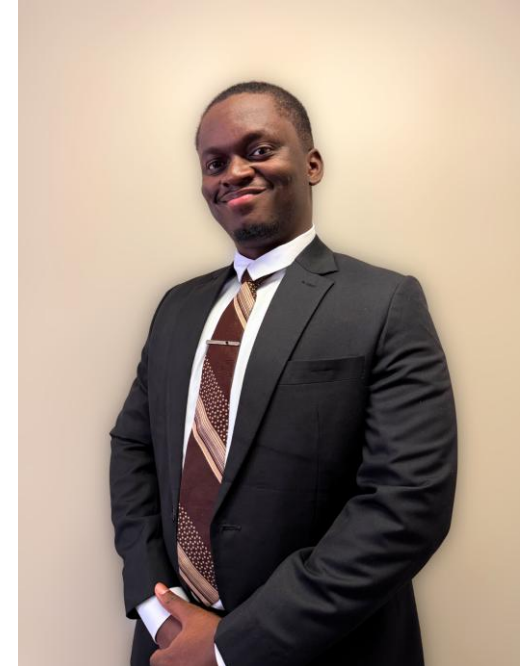
NATIONAL SOCIETY OF BLACK ENGINEERS

**Agents at Work – Accelerating and
Innovating Business Processes with
AI Agents**

- Hi! I am Ridwan Amure
- Researcher at COSMOS Research Center
- Area of interest: Machine Learning, Network Science, Applied AI
- In 2024, we prompted AI. In 2025, we automated with AI. In 2026, we orchestrate AI.



SCAN ME



What You Will Walk Away With

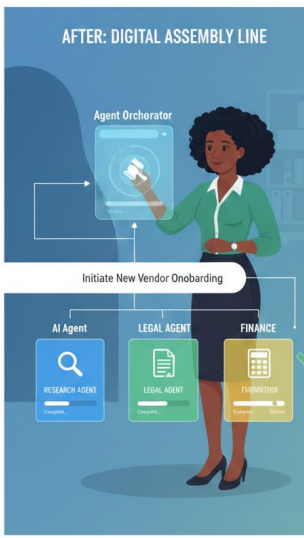
- Understand how AI agents differ from chatbots
- Design a multi-agent workflow for a business task
- See a working agent system in action
- Know how to build your own after today

The Efficiency Gap

- **Manual Labor:** Copy-pasting data between Email, Excel, and CRM.
- **Cognitive Load:** 20+ minutes to "re-focus" after every interruption.
- **Fragility:** If one person is out, the entire process stops.



The Efficiency Gap



- **Single Intent:** One high-level command (e.g., "Onboard this vendor").
- **Parallel Execution:** Research, Legal, and Finance agents working simultaneously.
- **Scalability:** The system handles 1 or 100 requests with the same speed.

The Anatomy of an AI Agent

- The Brain:
- How the agent breaks a big goal into steps
- Example: To pay this invoice, I first need to verify the tax ID
- The Hands:
- How the agent interacts with your business world
- Example: Accessing Gmail, SAP, Slack, or a SQL Database
- The Memory
- How the agent remembers what happened 5 minutes ago.
- Example: Recalling that a vendor was already flagged for a delay.



Logic and Planning



Action and Execution



Experience & State

Agents vs. Traditional Automation

Feature	Traditional Automation (Scripts/RPA)	AI Agents (The New Way)
Handling Input	Requires clean, structured data (CSV/DB).	Handles messy data (Emails, Voice, PDFs).
Logic	Rigid "If-This-Then-That" chains.	Adaptive "Reasoning Loops" (ReAct).
When it Breaks	Stops and errors if a UI button moves.	Re-plans and finds a new path to the goal.
Main Effort	Maintenance & Debugging.	Orchestration & Governance.



What is NOT an AI Agent

✗ Not
replacing
engineers

✗ Not
AutoGPT
running wild

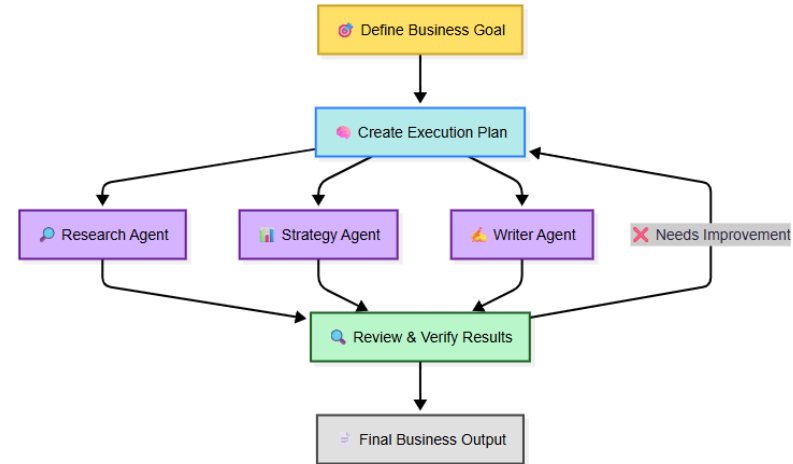
✗ Not just a
prompt

✓ A controlled
decision workflow

✓ Just
another tool in
your tool box

Agent Lifecycle (Execution View)

This is what happens every time you hit 'run'



Live Demo Rules

- Agents are imperfect
- We care about process, not perfect answers
- Logs matter more than output



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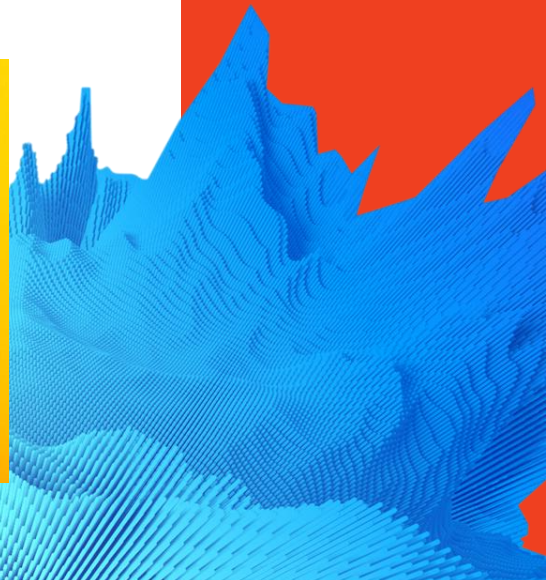
Demo 1: The Strategic Research Swarm

- Agent 1: The Researcher (Role: Senior Market Research Analyst)
 - Task: Find the top 3 competitors and their pricing models.
- Agent 2: The Strategist (Role: Business Growth Consultant)
 - Task: Analyze the research and find one "Gap" or "Weakness" for each competitor.
- Agent 3: The Writer (Role: Chief Communications Officer)
 - Task: Draft a 1-page executive summary memo for the CEO.



SCAN ME

What Just Happened



Demo 2: The Customer Support Triage Agent

- A company receives hundreds of customer messages daily.
- Today, humans manually read, classify, and route them.
- Let's see how agents handle this

Agent	Role
Classifier Agent	Categorizes issue (Billing, Bug, Account, Other)
Priority Agent	Assigns urgency (Low / Medium / High)
Routing Agent	Decides which team handles it
Supervisor Agent	Flags risky or ambiguous cases

Human-in-the-Loop & Guardrails

- Add Approval
- Add checkpoints
- Add Tool restrictions
- Add Audit logs
- Add Cost caps

How You'd Build This Yourself

Python + LLM

LangChain
N8n
CrewAI

Agent framework (or
custom loop)

Tools = APIs

Start with one agent
Build lean
Validate every step
Log activities



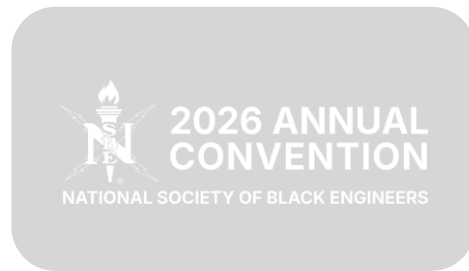
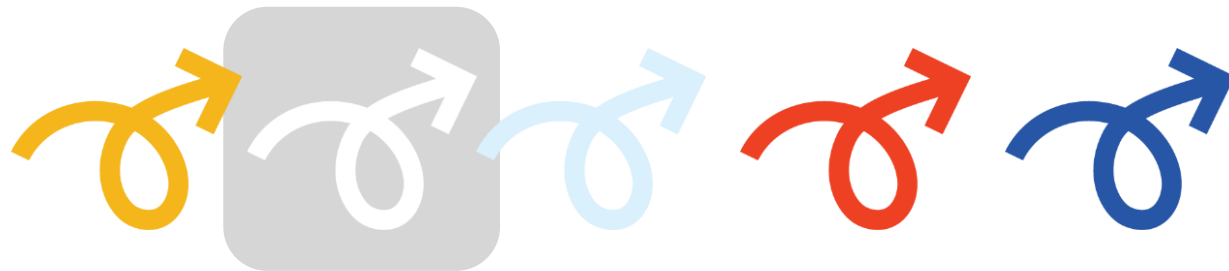
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THANK YOU



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